



Society of Petroleum Engineers

SPE-227758-MS

An Optimization Framework for Automatic History Matching of Aquifer Drive Reservoir with Geological and Geostatistical Data Integration

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227758-MS](https://doi.org/10.2118/227758-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Lately, we adapted two stochastic gradient algorithms, Adam and AdamW originally designed to train large deep learning models for automatic history matching of three-phase production data. In this paper, we extend their application by developing an optimization framework that systematically integrates well production data, geological models, and geostatistical constraints into reservoir simulation models. We theoretically show that the automatic history matching process can be formulated as a reaction–diffusion system. Solving this reaction–diffusion equation corresponds to a denoising diffusion process, where the reservoir parameters gradually evolve over iterations, reducing uncertainty while maintaining geological integrity. By adaptively adjusting governing coefficients, the proposed framework preserves geological structure while effectively reducing discrepancies between simulated and observed production data.

A key innovation in this work is the introduction of two complementary nonlinear diffusion formulations to guide spatial regularization. The first suppresses smoothing in regions where the optimizer identifies significant parameter variations, thereby preserving geologically meaningful features such as faults and facies boundaries. The second encourages structural alignment with a reference model by promoting diffusion where the estimated model deviates from the prior, without enforcing its absolute values. Together, these formulations provide a physically interpretable mechanism to jointly minimize production data mismatch while regularizing toward geologically consistent spatial structures during history matching. Model prediction is conducted using a three-dimensional, three-phase black oil simulator on a corner-point grid. Gradients are computed using an embedded adjoint method. These include not only grid permeability and porosity, but also aquifer properties such as permeability, porosity, thickness, and radius—treated as unknowns and estimated automatically.

Our case studies demonstrate three key advantages. First, the framework preserves sharp geological features—such as facies boundaries and low-permeability barriers—by employing nonlinear diffusion strategies that either suppress smoothing in mismatched regions or extract structural alignment from prior models. Second, it integrates geostatistical constraints into the optimization, ensuring consistency with spatial priors. Third, it improves convergence stability by iteratively updating all parameters—including aquifer properties—within a reaction–diffusion formulation guided by adjoint gradients, allowing for efficient and robust history matching over long production periods. These results show that the proposed

approach provides a physically interpretable mechanism to jointly minimize production data mismatch while regularizing toward geologically consistent spatial structures during history matching.

Introduction

Accurate reservoir characterization requires the integration of dynamic production data with prior geological and geostatistical information. While production history provides essential temporal constraints, it is typically sparse and spatially indirect. In contrast, geological data—such as depositional environments, facies distributions and fault structures—and geostatistical models derived from well logs and seismic interpretations offer high spatial resolution but lack direct temporal alignment. Reconciling these heterogeneous data sources within a unified modeling framework remains a central challenge in reservoir engineering.

Automatic history matching seeks to reduce the mismatch between observed and simulated production data by adjusting spatially distributed parameters such as permeability. This inverse problem is ill-posed and commonly stabilized through regularization, including Tikhonov (L2-norm) and Total Variation (TV) techniques that penalize rapid spatial variation (Engl et al., 1996). However, such methods often oversmooth the model and obscure key geological features like faults, streaks, and facies boundaries.

Geostatistical prior information can be rigorously incorporated through a Bayesian framework, in which spatial correlation structures are represented by a prior covariance matrix (Tarantola, 2005). This ensures that model updates are consistent with known geological patterns. Early applications of this approach involved conditioning permeability fields to well-test and production data using sensitivity-based methods (Chu et al., 1995; He et al., 1997; Wu et al., 1999). More recently, ensemble-based methods such as the Ensemble Kalman Filter (EnKF) introduced localization strategies to reduce spurious long-range correlations and computational cost (Emerick and Reynolds, 2011). Nonetheless, the need to invert or manipulate large covariance matrices remains a computational barrier for field-scale applications.

Several techniques have been proposed in the SPE literature to improve geological realism during history matching. Hu (2000) introduced the Gradual Deformation Method (GDM), which allows continuous deformation of geostatistical models while preserving their statistical properties. Caers and Hoffman (2002) developed a training image-based MCMC approach to retain geological patterns during inversion. Ravalec and Hu (2004) combined pluri-Gaussian simulations with assisted history matching to preserve channelized features. Jafarpour and McLaughlin (2008) utilized resampling-based updates within constrained prior spaces to maintain facies continuity. Wen and Chen (2002) used a geostatistical parameterization aligned with streamlines to focus updates on flow-sensitive zones.

In our earlier work (Wu, 2025a), we adapted a stochastic optimization algorithm—Adam—for use in automatic history matching. The initial parameter fields were constructed by assigning the prior mean value to each geological layer. This initialization strategy led to rapid convergence and preserved some large-scale geostatistical trends. However, the resulting models were overly smooth and lacked fine-scale geological variability. While effective for stabilizing the optimization process, this approach suppressed the heterogeneity required to capture key geological features such as streaks, faults, or facies boundaries.

To mitigate this, we included the stochastic realization as a soft prior constraint in the form of a regularization term proportional to $(u - u_{\text{prior}})/c$, where u_{prior} is the generated stochastic realization and c controls its strength (Wu, 2025b). This regularization stabilized early iterations by guiding the solution toward geologically plausible structures. However, we found that if the influence of this prior remained constant, the optimization process could stagnate or even increase the objective function—especially when the prior differed significantly from the true reservoir field. To tackle this problem, we progressively decayed the weight of the regularization term over time, allowing the optimizer to prioritize production data while still benefiting from early-stage structural guidance. However, as the influence of the regularization term

faded, the final estimated parameter fields became smoother—retaining some heterogeneity and large-scale geological trends but still lacking fine-scale geological features.

Notably, using the full prior realization directly as the initial guess often led to poor production data matches. This is because stochastic geostatistical simulation (SGS) methods honor global spatial relationships, but do not ensure accurate local parameter values around wellbores, which are critical for reproducing observed production behavior. As a result, even though the initial models were geologically plausible, they often failed to capture the detailed dynamics necessary for history matching.

In this paper, we propose a new framework for automatic history matching based on an edge-aware, orientation-sensitive reaction–diffusion system. The reaction term is defined as the variational gradient of the data misfit functional and serves as the primary driver to minimize the history matching objective. The diffusion term enforces geological regularity through spatially adaptive smoothing. Crucially, the diffusion is applied in an edge-preserving manner that suppresses smoothing across high-gradient regions, allowing key geological features—such as faults, facies boundaries, and sharp permeability contrasts—to be retained throughout model updates.

Structure-Preserving History Matching

To guide automatic history matching toward geologically consistent solutions, we define an energy functional $E(u)$ that reflects both the dynamic behavior of the reservoir and the spatial organization of subsurface properties. This energy functional consists of two components: a data misfit term and a regularization term:

$$E(u) = J_0(u) + \lambda \int_{\Omega} \phi(\|\nabla u\|) dx \quad (1)$$

where, $J_0(u)$ quantifies the misfit to observed production data and the second term enforces geological realism by penalizing undesirable spatial fluctuations. Data. The function $\phi(\|\nabla u\|)$ is a regularization function that controls the degree of smoothing, common choices include $\phi(r) = r^2$ (Tikhonov) and $\phi(r) = \sqrt{\epsilon + r^2}$ which is a Total Variation–like (TV-like) regularization. The regularization weight λ controls the tradeoff between matching data and maintaining geological coherence.

This formulation reflects the physical understanding that subsurface properties evolve through processes such as sedimentation, compaction, and diagenesis—tending toward spatially coherent states punctuated by sharp geological boundaries. Minimizing $E[u]$ is thus both a mathematical regularization strategy and a physically motivated model of subsurface evolution.

We simulate this evolution using a continuous gradient flow in artificial time:

$$\frac{\partial u}{\partial t} = -\frac{\delta E(u)}{\delta u} \approx -\frac{\delta J_0(u)}{\delta u} + \lambda \frac{\partial}{\partial u} \left(\int_{\Omega} \phi(\|\nabla u\|) dx \right) \quad (2)$$

This equation describes the steepest descent in the energy landscape defined by $E(u)$, guiding the parameter field $u(x, t)$ toward configurations that minimize data misfit while maintaining spatial structure. This is a gradient descent flow, which evolves u in artificial time t to minimize $E(u)$. The negative sign here is because we are moving in the direction of steepest descent.

The first term, $\frac{\delta J_0(u)}{\delta u}$ is computed using the adjoint-state method and provides the direction in which the model should change to reduce production misfit (Wu 2025a and Wu 2025b). The second term is derived by taking the variation of the regularization functional with respect to the model field. For a general smoothing function $\phi(\|\nabla u\|)$, the variational derivative is:

$$\frac{\delta}{\delta u} \left(\int_{\Omega} \phi(\|\nabla u\|) dx \right) = -\nabla \cdot \left(\phi(\|\nabla u\|) \frac{\nabla u}{(\|\nabla u\|)} \right) \quad (3)$$

Defining a nonlinear diffusion coefficient as:

$$g(\|\nabla u\|) = \left(\frac{\phi(\|\nabla u\|)}{\|\nabla u\|} \right) \quad (4)$$

we obtain the final update equation:

$$\frac{\partial u}{\partial t} \approx -\frac{\delta J_0(u)}{\delta u} + \lambda \nabla \cdot (g(\|\nabla u\|) \nabla u) \quad (5)$$

This equation resembles a reaction–diffusion system: the reaction term enforces data consistency, while the diffusion term performs structure-aware smoothing. In geologically uniform regions, diffusion proceeds freely; near sharp transitions such as faults or facies boundaries, the coefficient $g(\|\nabla u\|)$ decreases, suppressing smoothing and preserving key geological features. Unlike standard Laplacian regularization, this approach applies to a spatially adaptive diffusion mechanism informed by the local gradient structure. It offers a physically interpretable framework that integrates production data with geological realism, enabling robust and geologically consistent updates to reservoir parameters during optimization.

Nonlinear Diffusion for Edge Preservation

Conventional regularization methods often fail to distinguish between smooth zones and sharp boundaries, resulting in overly homogenized fields. To overcome this limitation, we introduce a nonlinear diffusion approach that selectively reduces smoothing across high-gradient regions, enabling edge preservation during history matching.

Our strategy is based on the edge-aware diffusion framework originally proposed by [Perona and Malik \(1990\)](#) in image processing. Their method adjusts diffusion strength according to the magnitude of the local gradient: full diffusion is applied in homogeneous regions where gradients are small, while diffusion is suppressed near discontinuities. This concept is particularly well suited to reservoir models, where abrupt spatial changes may correspond to physically meaningful transitions such as facies boundaries or faults.

We implement this behavior by defining a nonlinear diffusion coefficient as a function of the gradient magnitude:

$$g(r) = \frac{1}{\left[1 + \left(\frac{r}{\lambda_0} \right)^2 \right]} \quad (6)$$

where $r = \|\nabla u\|$ is the local gradient and λ_0 is a threshold parameter controlling edge sensitivity. In this formulation, the diffusion coefficient g determines how much smoothing is applied locally: When $g(r) \approx 1$, diffusion is strong. This occurs in regions where the gradient r is small—that is, where the model is relatively flat or homogeneous. In these zones, smoothing helps reduce noise or irregularities without harming geological realism. When $g(r) \ll 1$, diffusion is suppressed. This happens near sharp transitions where r is large—such as facies boundaries, streaks, or faults. Reducing $g(r)$ in these regions helps prevent smoothing across high-gradient geological features, which is crucial for preserving structure during model updates.

This edge-aware mechanism ensures that the diffusion term enforces regularity only where appropriate. If we did not suppress diffusion near discontinuities (i.e., if g remained high even at large gradients), important geological contrasts would be blurred out over iterations. Therefore, preserving streaks and faults requires designing $g(r)$ such that it decays rapidly as gradient magnitude increases.

[Fig. 1](#) illustrates the behavior of $g(r)$ and its derivative. The left panel shows how diffusion strength decreases with increasing gradient magnitude, while the right panel highlights how this suppression rate reaches a maximum near $r = \lambda_0$ and then flattens. This design helps preserve geological boundaries while still allowing regularization in less sensitive areas.

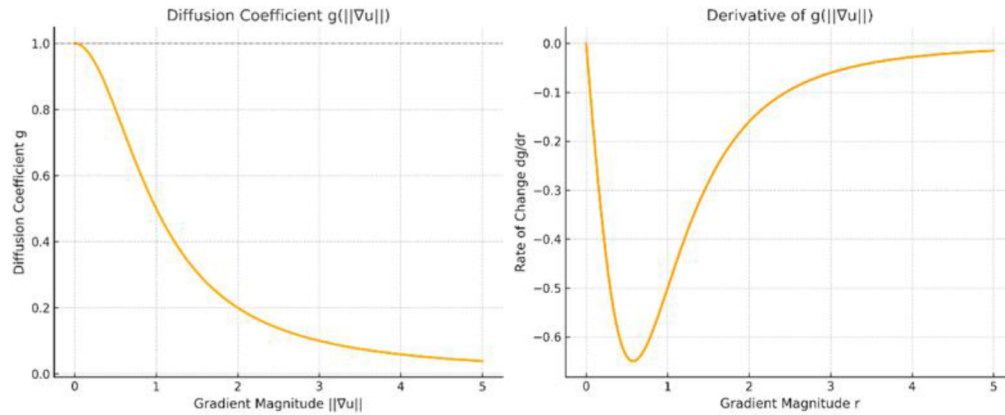


Figure 1—(Left) Nonlinear diffusion coefficient $g(r)$ based on gradient magnitude. Diffusion is strong in smooth regions and suppressed near sharp geological features. (Right) The rate of change dg/dr shows how quickly diffusion strength drops near structural boundaries.

This adaptive diffusion behavior is essential for maintaining high-contrast features throughout the iterative history matching process. It prevents critical geological structures from being eroded by uniform smoothing while supporting numerical stability and model convergence.

Reference-Guided Diffusion for Spatial Structure Alignment

In our previous work (Wu, 2025a; Wu, 2025b), we used the layer mean as a default initialization when applying Adam-based optimization for automatic history matching. This choice typically led to better convergence behavior, but the resulting models were often overly smooth and lacked geological realism. Alternatively, initializing the model using sequential Gaussian simulation (SGS) introduces spatial variability consistent with geostatistical priors. However, SGS-based initializations frequently result in poor convergence. Even when convergence is achieved, the final estimated model remains strongly biased toward the initial realization, with the estimated parameters closely resembling the SGS initialization. This highlights the challenge of reconciling prior structural patterns with production-driven data updates in the optimization process. To address this challenge, we propose a diffusion mechanism that extracts spatial structure from a geostatistical reference model without enforcing its absolute values. We define a spatial misalignment metric based on local deviations between the estimated parameter field $u(i, j, k)$ and a reference realization $v(i, j, k)$. Specifically, we compute

$$r(i, j, k) = \sqrt{\sum_{\text{neighbours}} (u_{nb} - v(i, j, k))^2} \quad (7)$$

Here u_{nb} denotes values of u at neighboring grid points. This approach shifts the focus from matching the absolute values of the prior model to aligning with its spatial structure. When the estimated model $u(i, j, k)$ locally resembles the reference $v(i, j, k)$, the misalignment metric $r(i, j, k)$ is small, and the diffusion coefficient $g(i, j, k) \approx 1$, encouraging smoothing to reinforce this agreement. In contrast, when the algorithm detects large deviations—such as at faults, streaks, or well-affected regions— $r(i, j, k)$ becomes large. By choosing a small threshold λ_0 , the diffusion coefficient $g(i, j, k)$ becomes very small, effectively preserving these locally significant variations. This design intentionally suppresses smoothing in regions where the optimizer has identified important mismatches, enabling dynamic data to override misleading prior structures. As a result, the model adapts to data while still benefiting from structural guidance encoded in the reference realization.

To examine how different formulations of the diffusion coefficient influence history matching, we compare two strategies based on the misalignment metric defined in Eq. (7), which quantifies the local structural deviation between the estimated model and a reference realization. The first formulation, given in

Eq. (6), corresponds to an edge-preserving diffusion scheme. It suppresses smoothing in regions with large misalignment (e.g., near faults or well-affected zones), enabling the model to deviate from the prior where necessary. In contrast, the second formulation, $g(r) = 1 / \left[1 + \left(\frac{\lambda_0}{r} \right)^2 \right]$ promotes stronger diffusion in regions with large mismatch, thereby encouraging the estimated model to align more closely with the structural patterns encoded in the reference model. When r is small—indicating strong local agreement with the prior—diffusion is reduced to preserve these features. The behavior of this formulation is controlled by the threshold parameter λ_0 . When $\lambda_0 = 0$, the diffusion coefficient becomes 1 everywhere, resulting in strong isotropic smoothing across the entire domain. Conversely, when $\lambda_0 \rightarrow \infty$, the diffusion coefficient $g(r) \rightarrow 0$ uniformly, effectively deactivating the diffusion term. In this limit, the algorithm reduces to the standard Adam-based optimization without any spatial regularization. These two formulations represent complementary philosophies: the first emphasizes preserving features discovered during optimization, while the second actively enforces spatial coherence with the prior.

Case Study

The PUNQ-S3 reservoir model is defined on a corner-point grid with 1,761 active blocks across 19×28 cells in the horizontal plane and 5 vertical layers. Each block spans approximately $590.4 \text{ ft} \times 590.4 \text{ ft}$, with slight tilting to conform to structural geometry. The reservoir features a strong edge aquifer to the west and north, faults along the east and south boundaries, and a small central gas cap situated in a domed structure. Well locations and grid elevations are shown in Fig. 2. Fig. 3 and Fig. 4 present the permeability and porosity distribution across the five layers used in the truth case.

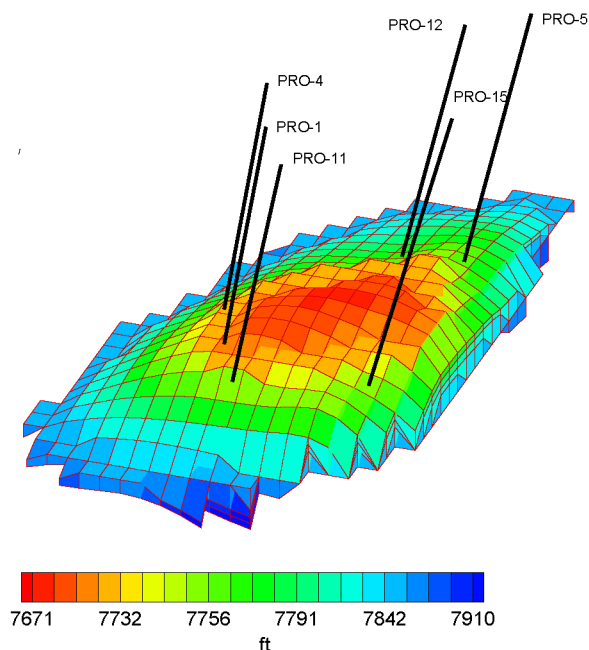


Figure 2—Elevation of the corner point grids with six wells located in the grids.

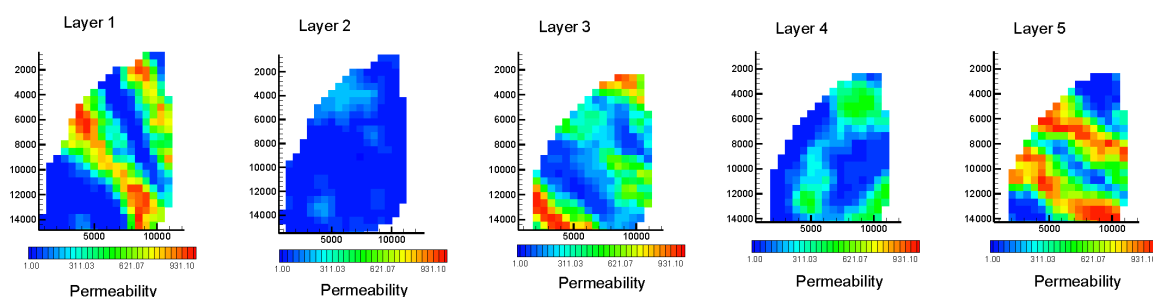


Figure 3—True horizontal permeability fields

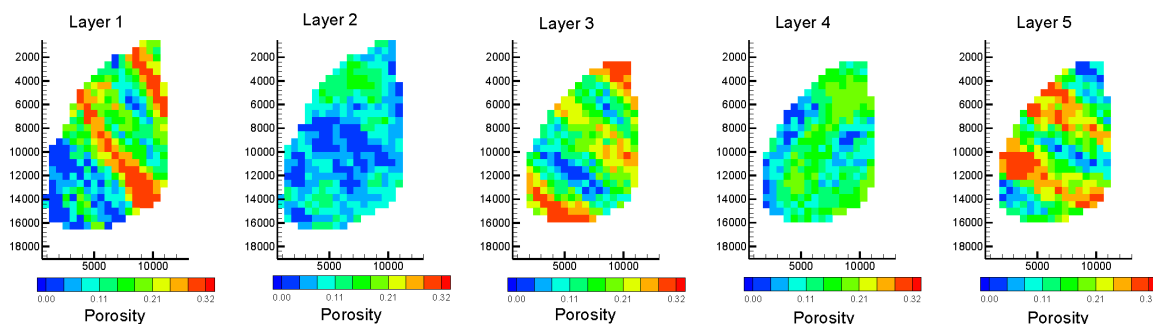


Figure 4—True porosity fields

The simulation period includes one year of extended well testing, three years of shut-in, and 16.5 years of field production. Each well is tested annually for two weeks. Aquifer support is sufficient to maintain pressure, and water injections are not required. Notably, well 11 experienced water breakthroughs in the seventh year, while wells 1 and 4 began producing gas in the fourth and fifth years, respectively. Production constraints included a constant oil rate of 943.47 STB/day, with a switch to constant BHP control once pressures dropped below 1740.46 psi. The only prior information about the reservoir parameters consists of the mean values of permeability and porosity, along with their standard deviations, as shown in Table 1.

Table 1—Summary of means and standard deviations of permeability and porosity for each layer in the true case.

Property	Horizontal Permeability (md)		Vertical Permeability (md)		Porosity	
Statistical Parameters	Mean	St Dev	Mean	Std Dev	Mean	St Dev
Lay 1	374.71	134.28	164.18	21.67	0.16	0.031
Lay 2	44.06	26.19	12.66	9.13	0.080	0.013
Lay 3	397.03	117.45	197.75	31.47	0.180	0.024
Lay 4	92.64	62.29	55.52	15.19	0.17	0.018
Lay 5	504.93	134.37	242.99	24.85	0.19	0.028

The goal of history matching is to estimate the spatial distribution of permeability and porosity by minimizing mismatch in water cut (over 16.5 years). These two properties are treated as independent across grid blocks, with no assumed correlation. The aquifer model parameters are considered as unknown and included into the iteration of the history matching. Further model details are provided in Barker et al. (2001), Gu and Oliver (2005), and de Santana et al. (2008).

Adaptive Diffusion for Feature Preservation

In this case, the reservoir model is initialized using a Gaussian distribution with mean equal to the layer average and a small fraction (e.g., 0.1) of the layer standard deviation. To maintain variability and avoid premature convergence to oversmoothed solutions, small Gaussian noise $0.01 * \sigma$ is injected at each iteration. The diffusion coefficient is defined as which suppresses smoothing in regions where the current model diverges from the reference. This edge-preserving formulation is designed to allow the optimizer to identify and retain sharp geological contrasts.

Fig. 7 and Fig. 8 show the estimated horizontal permeability and porosity fields across all five reservoir layers for the feature-preserving diffusion scheme. In this case, the initial model was generated from a purely stochastic Gaussian realization using $0.1 \times$ the standard deviation. Despite this weak initialization, the final estimates display spatial trends that are partially consistent with the reference model—particularly in layers 1, 3, and 5, which correspond to perforated production intervals. Notably, similarities are observed near Well 4, Well 11, and Well 12, where the estimated fields align with structural patterns present in the reference permeability and porosity fields (Figs. 5 and 6). These results suggest that the adaptive diffusion scheme can recover meaningful features during optimization, even when starting from a low-variability prior. The diffusion coefficient in this scheme is computed from the misalignment metric in Equation 6, which reduces smoothing in regions where structural gradients are high. However, due to the small initial variability (0.1σ), the smoothing effect remains relatively strong, and the resulting models—while structurally meaningful—tend to be smoother than the reference.

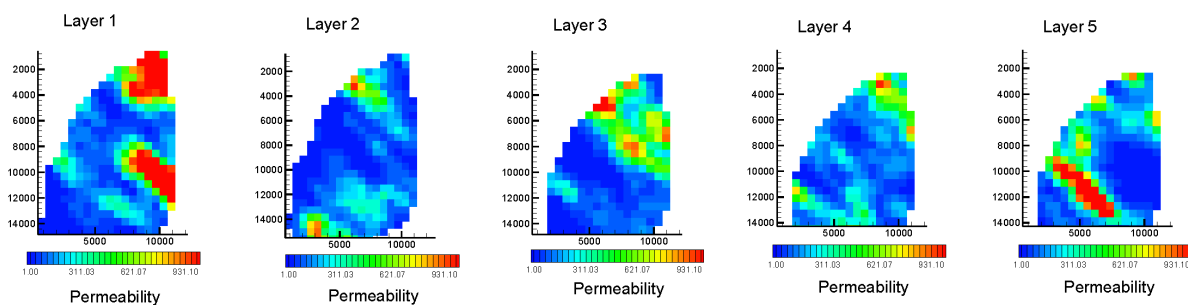


Figure 5—Reference horizontal permeability fields.

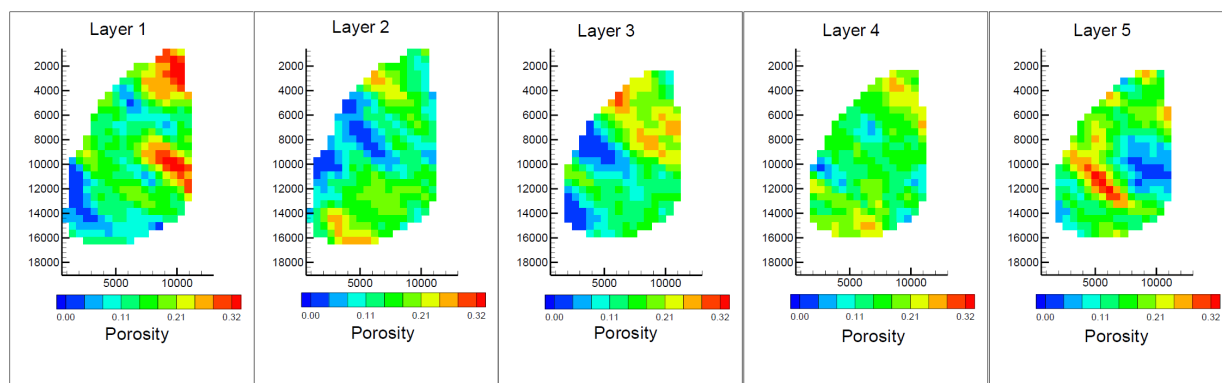


Figure 6—Reference porosity fields.

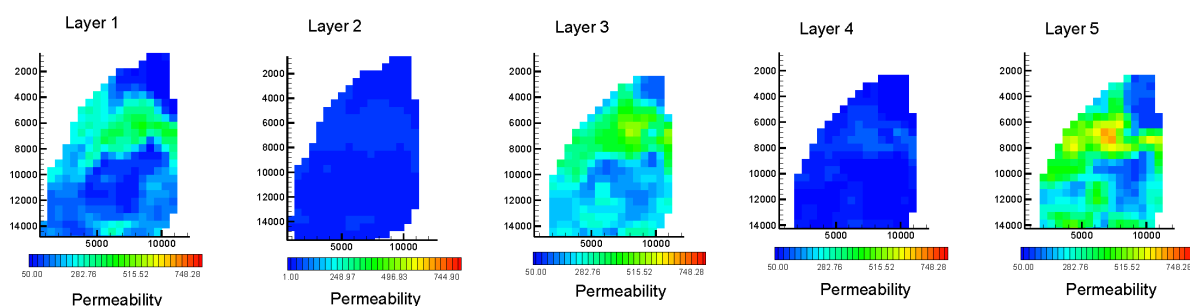


Figure 7—Estimated horizontal permeability using $0.1\sigma(\text{layer})$ Initialization and reference-guided diffusion with $\lambda_0 = \sigma(\text{layer})$

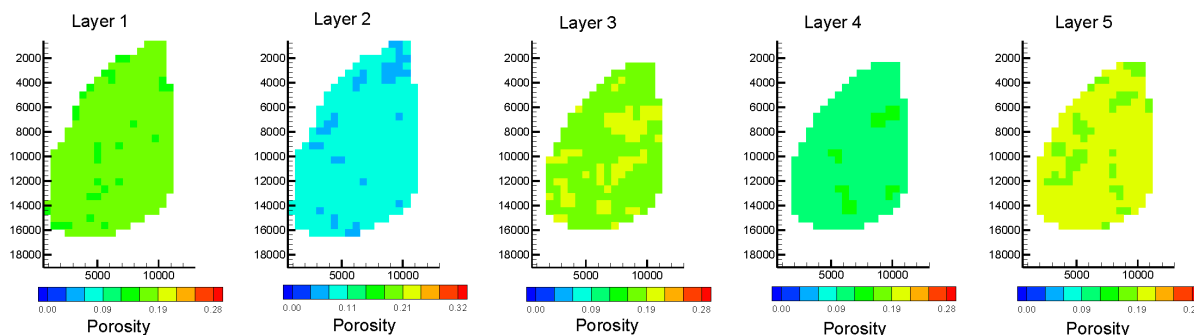


Figure 8—Estimated porosity using $0.1\sigma(\text{layer})$ initialization and reference-guided diffusion with $\lambda_0 = \sigma(\text{layer})$

Fig. 9 and Fig. 10 present results for a case with higher initial variability (0.8σ) and the same feature-preserving diffusion formulation. Although the initialization is a pure Gaussian realization without imposed geological structure, the increased variability introduces broader spatial fluctuations. This roughness allows the diffusion mechanism to preserve and enhance contrasts during optimization. As a result, the final permeability and porosity fields retain more heterogeneity and exhibit sharper features. Compared to the reference models (Figs. 5 and 6), these results show improved alignment of high-permeability zones and variations in the parameter fields, particularly in the lower layers, confirming that the combination of rough initial conditions and adaptive diffusion helps recover meaningful spatial patterns.

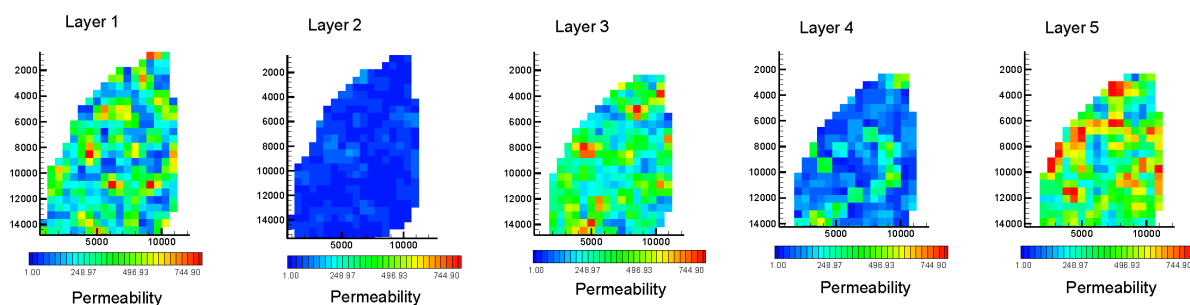


Figure 9—Estimated horizontal permeability using $0.8\sigma(\text{layer})$ Initialization and reference-guided diffusion with $\lambda_0 = \sigma(\text{layer})$

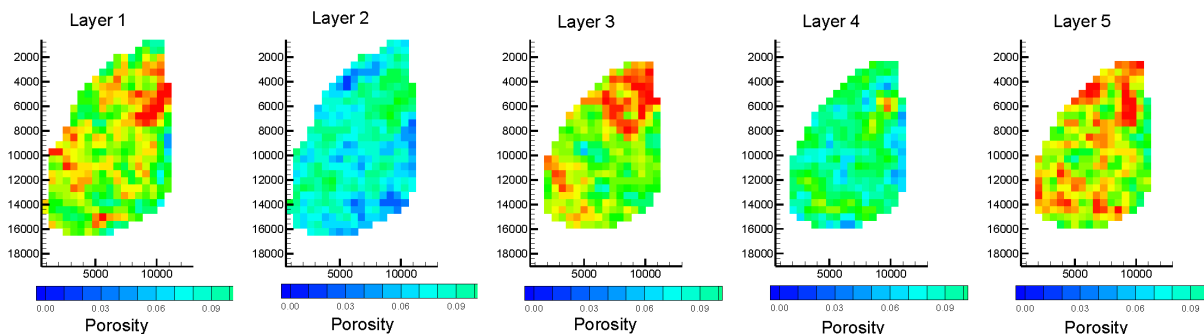


Figure 10—Estimated porosity using $0.8\sigma(\text{layer})$ Initialization and reference-guided diffusion with $\lambda_0 = \sigma(\text{layer})$

Structure-Preserving Diffusion Guided by Prior Alignment

In this case study, the reservoir model is initialized using a Gaussian realization with a relatively large standard deviation (0.8σ per layer), introducing strong heterogeneity from the beginning. The diffusion coefficient follows the inverse formulation described in Equation (6), where diffusion is promoted in regions with high misalignment between the estimated and reference models. This strategy actively encourages alignment with the spatial patterns embedded in the reference realization, especially in areas where the estimated field significantly deviates from prior structure.

Unlike the feature-preserving scheme, which suppresses smoothing near geological boundaries, this reference-guided smoothing formulation enforces spatial coherence by diffusing more strongly in high-mismatch areas. This helps to stabilize updates, particularly when the initial model is rough or contains spurious noise. Importantly, the framework honors the structural layout of the reference realization without imposing its exact numerical values, allowing local adjustments guided by production data.

Fig. 11 and Fig. 12 show the estimated horizontal permeability and porosity fields obtained using this approach. The results demonstrate structural similarity with the true parameter fields, including the reconstruction of high- and low-contrast zones across layers. The estimated fields closely reflect the spatial trends encoded in the reference models (cf. Figs. 5 and 6). These results highlight several key advantages of the approach. The use of large initial variability provides sufficient structural information to guide the diffusion process. Moreover, the small noise injection ($0.01 \times \sigma$) helps prevent over-smoothing and promotes exploration of the parameter space, reducing the risk of early convergence to suboptimal solutions.

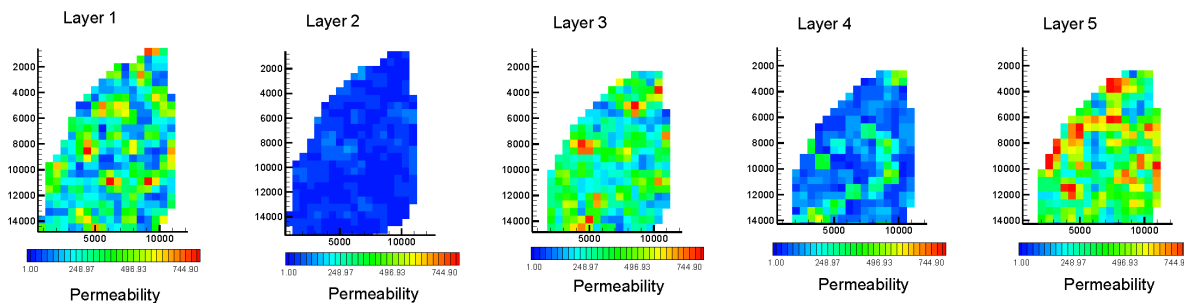


Figure 11—Estimated horizontal permeability using $0.8\sigma(\text{layer})$ Initialization and reference-guided diffusion with $\lambda_0 = \sigma(\text{layer})$

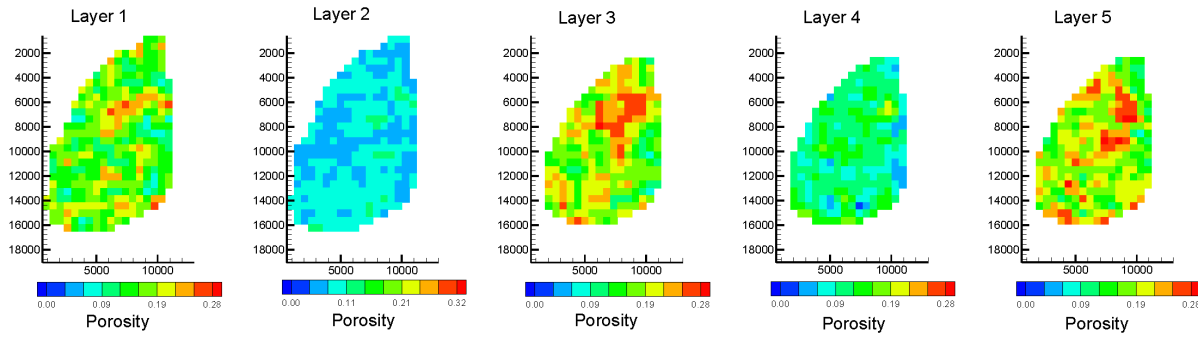


Figure 12—Estimated porosity using $0.8\sigma(\text{layer})$ Initialization and reference-guided diffusion with $\lambda_0 = \sigma(\text{layer})$

Fig. 13 and Fig. 14 display the initial horizontal permeability and porosity fields used in this case. These fields are generated from pure Gaussian realizations using the standard deviation of each layer (0.8σ), resulting in strong local heterogeneity without any imposed spatial structure or geological trends. This high level of variability provides a challenging starting point for history matching, as it lacks coherent patterns and can significantly influence the evolution of the model during optimization.

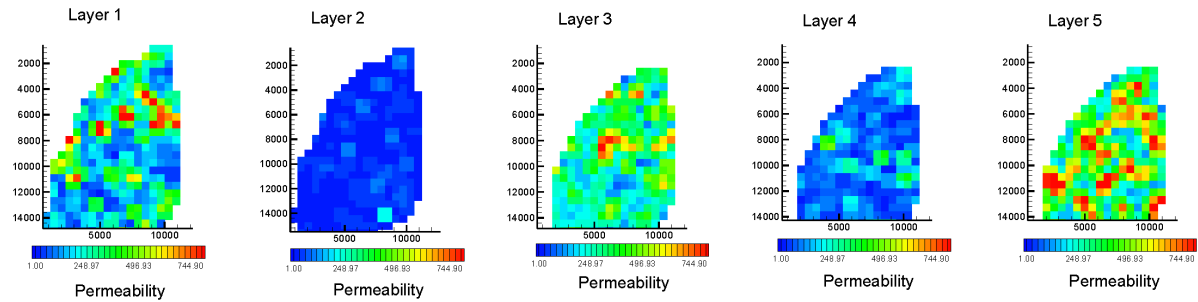


Figure 13—Initial horizontal permeability field generated from a pure Gaussian realization using $0.8\times$ the standard deviation per layer.

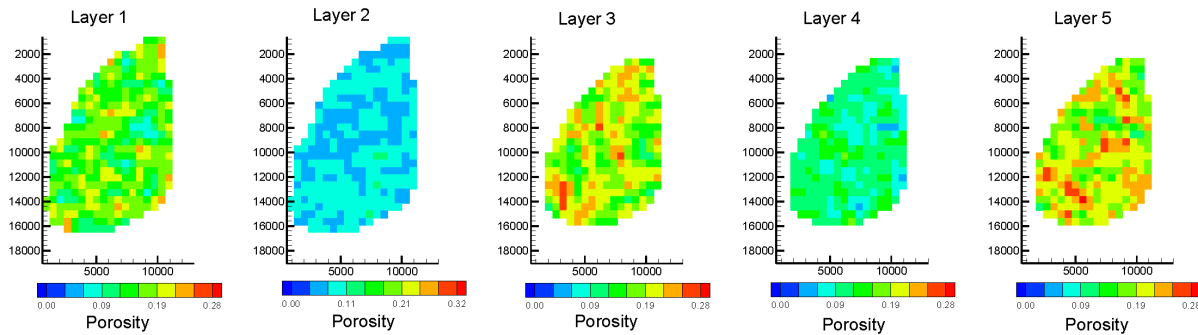


Figure 14—Initial porosity field generated from a pure Gaussian realization using $0.8\times$ the standard deviation per layer.

Fig. 15 and Fig. 16 show the corresponding final estimates obtained using a large diffusion strength, with the threshold parameter set to $\lambda_0 = 0$. In this configuration, the reference model plays no role in shaping the diffusion coefficient, meaning that the updates are governed solely by the initial realization and the production data misfit. When comparing the estimated fields to the initial ones, a structural resemblance emerges. Some of the spatial features seen in the final models reflect the variability present in the initial realization. This outcome suggests that the strong heterogeneity of the initial field can dominate the optimization trajectory—particularly in the absence of a structural prior—and may limit the ability of production data to restructure the parameter fields. While the reaction–diffusion mechanism helps

regularize the updates, it cannot fully override the bias introduced by highly variable initial conditions when $\lambda_0 = 0$.

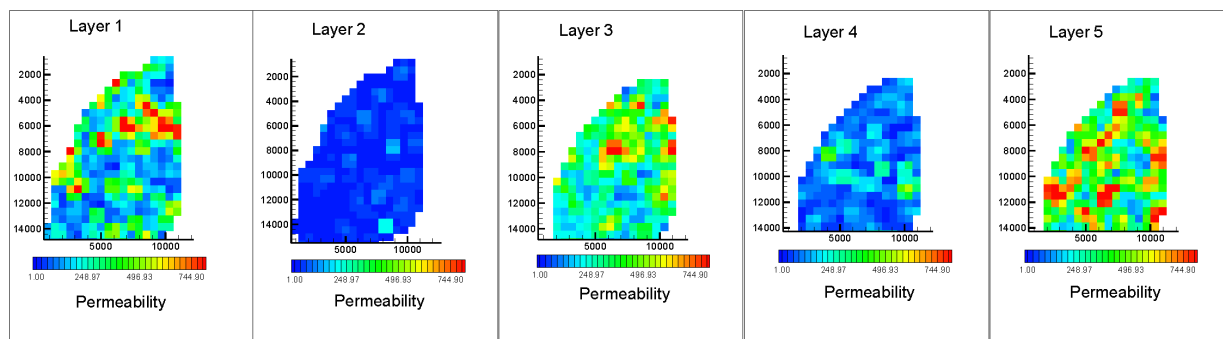


Figure 15—Estimated permeability using $0.8\sigma(\text{layer})$ Initialization and reference-guided diffusion with $\lambda_0 = 0$.

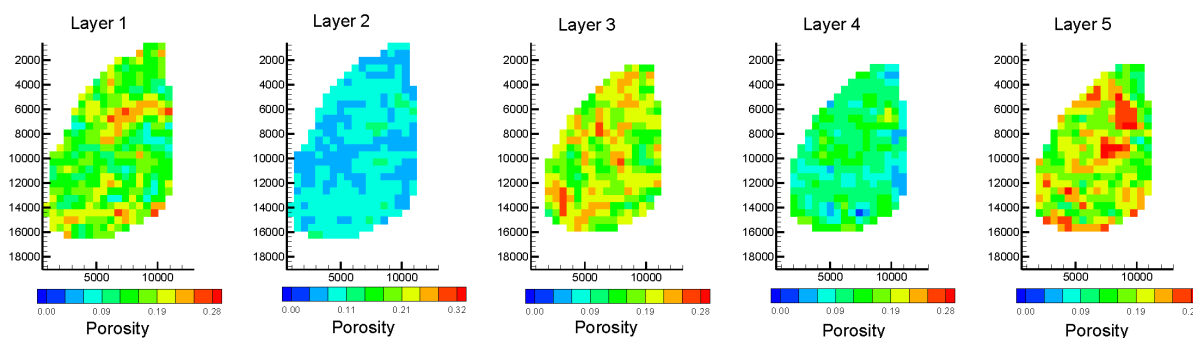


Figure 16—Estimated porosity using $0.8\sigma(\text{layer})$ Initialization and reference-guided diffusion with $\lambda_0 = 0$.

This case illustrates two important concerns. First, too rough initial models can pose risks of numerical instability, especially in multiphase simulations where sharp parameter contrasts may hinder convergence. Second, without reference alignment, the optimization process becomes more sensitive to the randomness of the initial model, which can reduce the effective influence of production data. The final estimates may still largely reflect the spatial layout of the initial guess rather than converging toward a geologically consistent reservoir models.

These observations highlight a critical trade-off in diffusion-based history matching. High variability in the initial parameter fields may help preserve contrast and spatial structure; however, overly rough initial models can dominate the history matching process. In such cases, the resulting estimates may become overly dependent on the initial realization rather than being driven by production data. Future work should explore adaptive strategies to balance initial heterogeneity with reference-based guidance, supporting stable history matching and maintaining consistency with the prior geostatistical model

Fig. 17 presents the cumulative aquifer water influx for both the initial reservoir model and the history-matched models. A large deviation is observed between the initial and true cases, emphasizing the significance of aquifer parameter estimation. The water influx from the two aquifers differs considerably. Aquifer 1 contributes less water to the reservoir compared to the true case. Aquifer 2 contributes more water than expected.

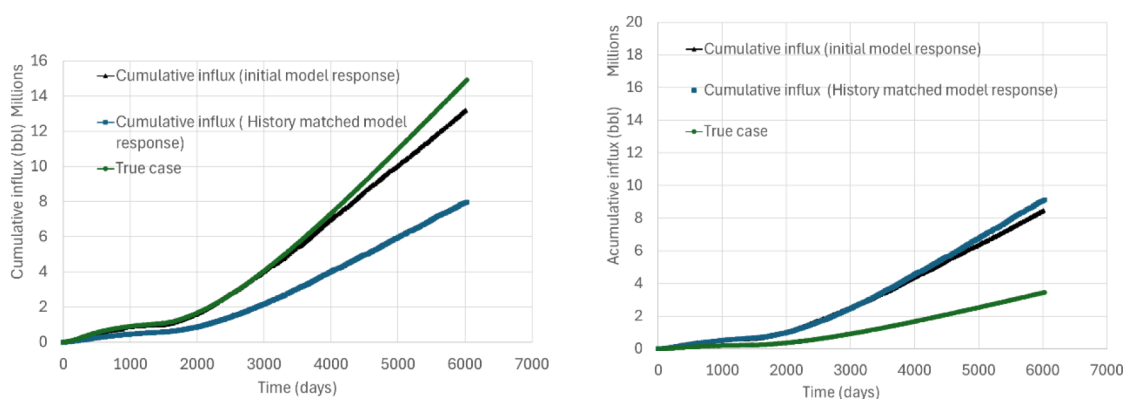


Figure 17—Comparison of true cumulative influx data, initial model prediction, and history matched model prediction.

These discrepancies highlight the high uncertainty associated with aquifer properties, reinforcing the necessity of incorporating aquifer parameters into history matching. The differences in water influx distributions suggest that incorrect aquifer properties can significantly distort reservoir pressure support and production predictions.

Table 2 summarizes the estimated aquifer properties after matching water cut data from four production wells. The results demonstrate the effectiveness of the proposed framework in jointly estimating both reservoir and aquifer parameters, a significant advancement over traditional history matching approaches that assume aquifer properties are known and fixed. However, despite reduced uncertainty, some estimated aquifer properties deviate significantly from the true values. This non-uniqueness in parameter estimation during history matching underscores the importance of integrating more observed data to improve the accuracy of aquifer property estimation.

Table 2—Comparison of Initial Guess, Estimated, and True Values for Aquifer Properties.

Aquifer	Property (md)	Initial guess	Estimated	True case
Aquifer 1	Permeability (md)	160.0	134.00	137.50
	porosity	0.16	0.27	0.21
	Radius (ft)	9184.00	10690.00	9840.00
	Thickness (ft)	46.00	90.00	64.30
Aquifer 2	Permeability (md)	160	342.00	137.50
	porosity	0.16	0.10	0.21
	Radius (ft)	9184.00	7856.00	10496.00
	Thickness (ft)	46.00	25.00	19.68

Figure 18 compares the observed water cut data, initial model predictions, and final history-matched results for Wells 4 and 5, which display contrasting behaviors. For Well 4, the observed data show water breakthrough occurring around day 4,300, yet the initial model prediction fails to capture any water production. In contrast, Well 5 exhibits delayed water breakthrough in the observed data—around 4,000 days—with a gradual increase in water cut. However, the initial model predicts an unrealistically early breakthrough around day 2,000 and a much steeper rise in water cut. After history matching, both wells show significantly improved alignment with the observed data. Notably, the optimization successfully corrected the substantial discrepancies in the initial predictions, despite the parameters being initialized without any well-specific tuning. This highlights the capability of the proposed reaction–diffusion framework to

reconstruct dynamic production behavior from mismatched initial models by incorporating production data effectively.

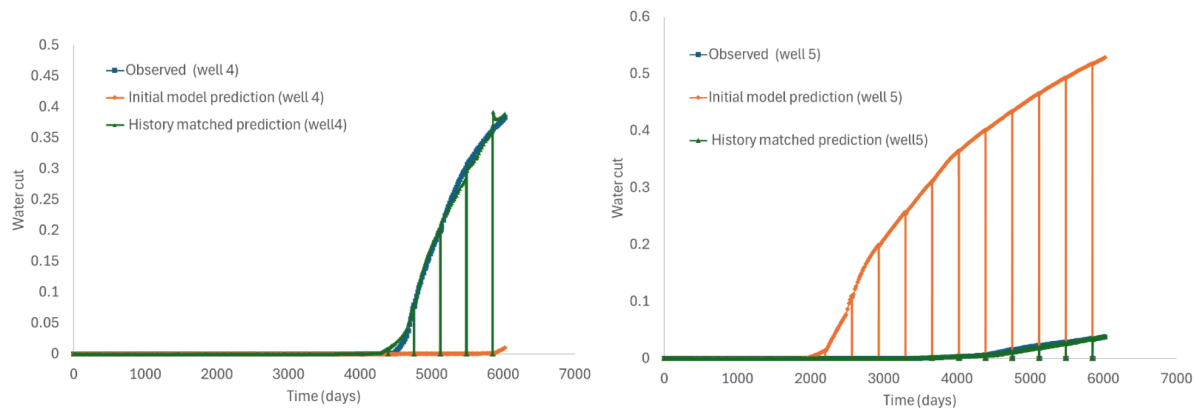


Figure 18—Comparison of observed water cut data, initial model prediction, and final history matching model prediction for Wells 4 and 5.

Figure 19 shows the normalized objective function decline over iterations. A characteristic zig-zag pattern is observed, particularly near the minimum. This behavior is primarily due to the use of a warm-restart learning rate schedule (Wu, 2025a), which periodically increases the learning rate. Additionally, the momentum term in AdamW contributes to this oscillatory pattern by introducing temporary overshooting as the optimizer nears a local minimum. While these mechanisms help the optimizer escape shallow minima and improve exploration, they also result in non-monotonic convergence. As the optimization progresses, the decline in the objective function gradually levels off—typical of gradient-based methods—indicating that further reductions become increasingly marginal. At this stage, convergence criteria can be defined based on flow rate misfit or relative changes in the objective function. To balance computational cost and model fidelity, a stopping condition should be determined when additional iterations yield diminishing returns in data fit.

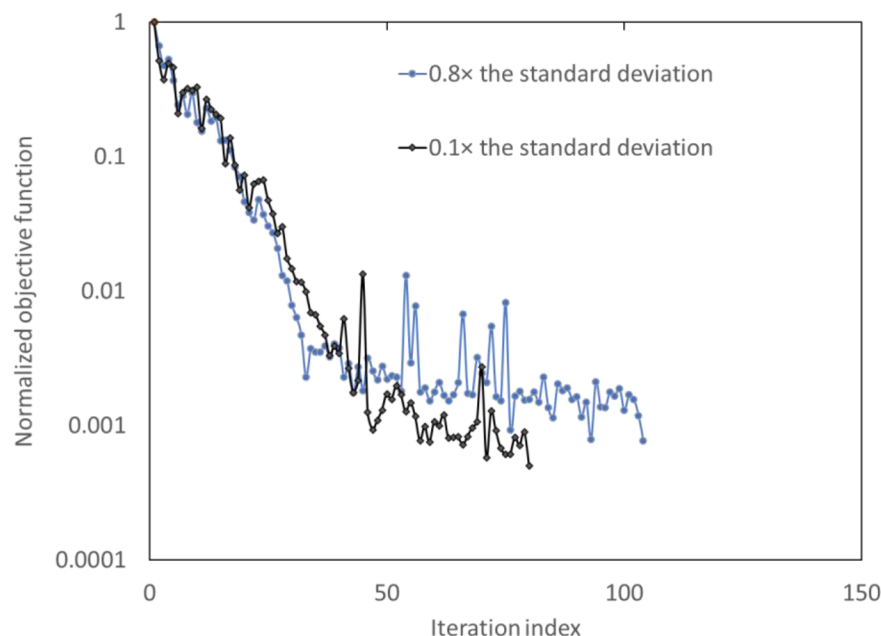


Figure 19—Objective function versus number of iterations for history matching.

Conclusions and Discussions

This paper presents a novel framework that integrates orientation-aware reaction–diffusion (RD) modeling with automatic history matching to improve the spatial coherence and convergence behavior of reservoir parameter estimation. By embedding nonlinear diffusion into the optimization loop, the framework integrates dynamic production data with spatial structure of prior geological knowledge into reservoir parameter estimation.

To mitigate the over-smoothing problem associated with layer-wise mean initialization, we introduced reference-guided diffusion, where the diffusion coefficient is a function of the spatial misalignment between the evolving estimate and a prior model. Two key formulations of the diffusion coefficient were investigated: one that suppresses diffusion in mismatched regions to preserve sharp features (edge-preserving), and another that enhances diffusion where structural misalignment is high to enforce alignment with the reference model. Both strategies successfully extracted geological structure from the prior model without enforcing absolute values, offering flexibility in adapting to dynamic data. A suite of case studies on the PUNQ-S3 reservoir benchmark demonstrated the effectiveness of the proposed framework. The results show that Gaussian initializations with partial standard deviation and mild noise injection encourage exploration and help avoid local minima but may slow convergence if the initial field is too rough.

The Orientation RD framework bridges geostatistical priors and dynamic constraints through a physically interpretable mechanism that balances structure preservation and production data integration. It provides a robust alternative to traditional regularization-based methods and provides new directions for history matching under geological uncertainty.

Relation to Existing Optimization Methods for History Matching

Steepest-descent with line-search, and ensemble-based methods—such as the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) and the Iterative Ensemble Smoother (IES) are well established in the history matching literature. These methods operate by perturbing prior model ensembles and updating them iteratively using Kalman-type formulas, often without requiring access to simulator source codes. While ensemble-based methods are widely used for uncertainty quantification, they typically require large ensemble sizes and may face challenges in strongly nonlinear systems such as three-phase flow. While BFGS is a well-known quasi-Newton method, it is less practical for three-phase history matching due to its high memory demands and reliance on line search. In contrast, AdamW offers low-memory, first-order updates with adaptive step sizes and does not require curvature approximations, making it more robust and efficient for high-dimensional, nonlinear problems with adjoint-computed gradients.

Our goal is not to replace these methods, but to show that the proposed reaction–diffusion framework offers a practical and technically robust alternative for three-phase history matching—one that adaptively regularizes parameter fields through spatially structured diffusion while preserving faults, facies boundaries, and sharp permeability contrasts during updates. This selective, orientation-aware smoothing mechanism provides geologically consistent model updates without the over-smoothing often observed in traditional regularization approaches, offering convergence stability and improved preservation of reservoir heterogeneity.

Scope with Respect to Non-Uniqueness

History matching problems admit multiple admissible solutions. A full exploration of this non-uniqueness—through Bayesian sampling or large ensembles—is beyond the scope of the present study. Instead, we begin from spatially uncorrelated Gaussian realization drawn from the prescribed layer means and standard deviations; no variogram structure is imposed at initialization. The required geological structure is introduced progressively during optimization: the reference-guided diffusion coefficient embeds prior spatial information at each iteration, so that channel trends, fault offsets, and facies boundaries emerge as the model evolves. In this way, the framework demonstrates that even from a structure-free starting point,

reaction–diffusion updates can recover a geologically plausible solution that honors both dynamic data and geological structure.

The PUNQ-S3 Reservoir Model

Although PUNQ-S3 has a relatively small number of grid cells, it retains critical complexities for history matching: layered geology, multiple sealing faults, a corner-point grid structure, strong edge-aquifer support, and a multiphase production history under a realistic operational schedule involving well shut-ins, rate/BHP switches, and phased production. These features make it a technically meaningful benchmark for evaluating the effectiveness of optimization methods. Rather than emphasizing model size which is often hardware-dependent, we highlight the number of simulator runs required to complete the entire history matching process as a more practical measure of computational cost. In this regard, the proposed reaction–diffusion framework, combined with the Adam optimizer, delivers geologically consistent parameter updates, fast convergence, and ease of implementation—all achieved with significantly fewer simulator calls compared to traditional gradient-based or ensemble-based methods.

Acknowledgements

The author gratefully acknowledges Dr. Shuiquan Li from the Enhanced Oil Recovery Institute, University of Wyoming, for sharing his valuable experience in reservoir simulation and modeling, particularly in history matching using commercial simulators.

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Appendix A

Discretization of the Structure-Preserving Reaction–Diffusion Model

The structure-preserving update equation derived in the main text is:

$$\frac{\partial u}{\partial t} \approx -\frac{\delta J_0(u)}{\delta u} + \lambda \nabla \cdot (g(\|\nabla u\|) \nabla u) + \epsilon \cdot \eta(x, y, z, t) \quad (\text{A} - 1)$$

We discretize this equation on a uniform 3D grid with spacings dx , dy , and dz , and use an implicit time integration scheme to advance the model in artificial time. ϵ controls the noise amplitude, often chosen as a small fraction (e.g., $0.01\times$ or $0.05\times$) of the prior layer's standard deviation.

Time Discretization (Implicit Scheme)

We discretize time using a backward Euler scheme:

$$\frac{u_{i,j,k}^{n+1} - u_{i,j,k}^n}{\Delta t^n} \approx -\frac{\delta J_0(u)}{\delta u} \Bigg|_{i,j,k}^n + \lambda \nabla \cdot (g^n \nabla u^{n+1})_{i,j,k} \quad (\text{A} - 2)$$

Here, the data misfit gradient $\frac{\delta J_0(u)}{\delta u}$ is evaluated at the previous time step n , and the nonlinear diffusion term is treated implicitly with coefficients g fixed from the previous state. This results in a linear system to be solved for u^{n+1} at each step.

Spatial Discretization

The diffusion operator $\nabla \cdot (g \nabla u)$ is discretized using central differences in all three spatial directions.

For each grid cell i, j, k , the contribution in the x-direction is:

$$\frac{1}{dx^2} \left[g_{i+\frac{1}{2},j,k}^{n+1} (u_i^{n+1} - u_i^n) - g_{i-\frac{1}{2},j,k}^{n+1} (u_i^{n+1} - u_{i-1}^n) \right] \quad (\text{A} - 3)$$

Similar expressions apply for the y- and z-directions, using spacings dy and dz . The midpoint diffusion coefficients are computed by averaging values at adjacent cells:

$$g_{i+\frac{1}{2},j,k}^{n+1} = \frac{1}{2} [g(\|\nabla u^n\|_{i+1,j,k}) + g(\|\nabla u^n\|_{i,j,k})] \quad (\text{A} - 4)$$

Gradient Magnitude Approximation

To evaluate the edge-preserving diffusion coefficient $g(\|\nabla u\|)$, we approximate the squared magnitude of the spatial gradient using centered finite differences from the previous time step u . This allows the reaction–diffusion update to remain explicit and linear, avoiding the need for nonlinear solvers.

$$\begin{aligned} \|\nabla u^n\|_{i,j,k}^2 = & \frac{1}{2} \left[\left(\frac{u_{i+1,j,k}^n - v_{i,j,k}^n}{dx} \right)^2 + \left(\frac{u_{i-1,j,k}^n - v_{i,j,k}^n}{dx} \right)^2 \right] + \frac{1}{2} \left[\left(\frac{u_{i,j+1,k}^n - v_{i,j,k}^n}{dy} \right)^2 + \left(\frac{u_{i,j-1,k}^n - v_{i,j,k}^n}{dy} \right)^2 \right] \\ & + \frac{1}{2} \left[\left(\frac{u_{i,j,k+1}^n - v_{i,j,k}^n}{dz} \right)^2 + \left(\frac{u_{i,j,k-1}^n - v_{i,j,k}^n}{dz} \right)^2 \right] \end{aligned} \quad (\text{A} - 5)$$

where (dx, dy, dz) are the grid dimensions. This expression is identical in form to a centered finite-difference gradient, except that each neighbor is compared with the reference value $v_{i,j,k}^n$ rather than the central estimate $u_{i,j,k}^n$. The resulting scalar $v_{i,j,k}^n$ drives the edge-aware diffusion coefficient

$$g(r) = \frac{1}{\left[1 + \left(\frac{r}{\lambda_0}\right)^2\right]} \quad \text{or} \quad (r) = \frac{1}{\left[1 + \left(\frac{\lambda_0}{r}\right)^2\right]}$$

depending on whether feature preservation or reference alignment. This approximation uses only known values at the previous time step, enabling an explicit update scheme that improves computational efficiency and numerical stability. The reference model $v(i,j,k)$ serves as the local baseline when evaluating spatial gradients, ensuring that the diffusion mechanism responds to structural deviations from the prior rather than absolute parameter values.